Local Cerebral Glucose Utilization in the Normal Conscious Macaque Monkey

C. Kennedy, MD, O. Sakurada, MD, M. Shinohara, MD, J. Jehle, and L. Sokoloff, MD

The [^{14}C] deoxyglucose method for quantitative determination of local cerebral glucose utilization was extended to the macaque monkey. The necessary constants required by its operational equation were evaluated. The lumped constant, measured in 7 normal conscious monkeys, was found to equal 0.344 (SEM, ± 0.036). The rate constants were also estimated and found to be very similar to those obtained previously in the rat.

With these essential constants evaluated, the method was applied to normal conscious monkeys. Local cerebral glucose utilization was found to vary markedly throughout the brain but to fall within two distributions, a higher one in gray matter and a lower one in white matter. In general, the values fell in a range to be expected from previous measurements of average energy metabolism in the brain as a whole. The values were considerably lower than those observed previously in the conscious rat.

Marked heterogeneity of rates of glucose utilization were observed in a number of anatomical structures. In some cases the patterns of heterogeneity were consistent with known histological cytoarchitecture; in others the heterogeneity did not conform with known cytoarchitectural features but corresponded to patterns previously demonstrated by electrophysiological techniques. Many of the regions of the cerebral cortex showed columnar patterns of distribution of higher and lower rates of glucose utilization. These may be a metabolic reflection of the columnar organization of function within the cerebral cortex.

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A method for quantitative simultaneous determination of the rates of glucose consumption in all of the macroscopic structural and functional components of the central nervous system has recently been reported by us [23]. The method employs 2-deoxy-D-[^{14}C] glucose ([^{14}C]DG) as a tracer for glucose metabolism and utilizes a quantitative autoradiographic technique to measure local carbon 14 concentration and to achieve spatial localization in specific components of the brain. The theoretical basis and operational design of the method are derived from analyses of a model based on the biochemical properties of 2-deoxyglucose in brain and the kinetics of the exchange of 2-deoxyglucose and glucose between plasma and brain tissue and their phosphorylation by hexokinase [23]. Mathematical analysis of this model leads to an operational equation that defines the local rate of glucose utilization in terms of measurable variables and definable constants. This equation is as follows:

$$R_i = \frac{C_i^*(T) - k_1^* e^{-(k_2^* + k_3^*)T} \int_0^T C_P^* e^{(k_2^* + k_3^*)t} dt}{\left[\frac{\lambda V_m^* K_m}{\Phi V_m K_m^*}\right]\left[\int_0^T (C_P^*/C_P)dt - e^{-(k_2^* + k_3^*)T} \int_0^T (C_P^*/C_P) e^{(k_2^* + k_3^*)t} dt\right]}$$

where R_i equals the rate of glucose utilization per unit mass of tissue, i; $C_i^*(T)$ equals the combined concentrations of [^{14}C]DG and [^{14}C]DG-6-phosphate in the tissue, i, at time, T, determined by quantitative autoradiography; C_P^* and C_P equal the arterial plasma concentrations of [^{14}C]DG and glucose, respectively; k_1^*, k_2^*, and k_3^* are the rate constants for transport of [^{14}C]DG from plasma to the tissue precursor pool, for its transport back from tissue to plasma, and for the phosphorylation of free [^{14}C]DG in the tissue, respectively; λ equals the ratio of the distribution space of [^{14}C]DG in the tissue to that of glucose; Φ equals the fraction of glucose which, once phosphorylated, continues down the glycolytic pathway; and K_m^*, V_m^*, K_m, and V_m are the familiar Michaelis-Menten kinetic constants of hexokinase for [^{14}C]DG and glucose, respectively.

In order to apply this equation to the measurement of local cerebral glucose utilization, a pulse of [^{14}C]DG is administered intravenously at zero time,

From the Laboratory of Cerebral Metabolism, National Institute of Mental Health, Department of Health, Education, and Welfare, US Public Health Service, Bethesda, MD, and the Department of Pediatrics, Georgetown University School of Medicine, Washington, DC.

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Address reprint requests to Dr Sokoloff, NIMH, 9000 Rockville Pike, Bldg 36, Room 1A-27, Bethesda, MD 20014.

and the time courses of the plasma [^{14}C]DG and glucose concentrations and the tissue ^{14}C concentrations at the time of killing are measured in each experiment. The rate constants k_1^*, k_2^*, and k_3^* and the lumped constant, which combines λ, Φ, V_m^*, V_m, K_m^*, and K_m into a single constant, are not measured at the same time. The values for these constants are determined in separate groups of animals by means of other procedures that have been previously described [23]. For reasons presented in the original report of the method [23], these constants can be expected to vary in different species of animals.

Thus far, the rate constants and the lumped constant have been evaluated only in the rat, and it has therefore been impossible to determine local cerebral glucose utilization quantitatively in any other species. There are, however, situations in which studies on more complex nervous systems than that of the rat are desirable. The monkey is a primate with a nervous system closer to that of man and is a more appropriate model for many neurophysiological, neuropharmacological, and neuropathological investigations. The present studies were carried out to extend the applicability of the [^{14}C]DG method to the monkey. The necessary constants have been determined and used to obtain normal values for local cerebral glucose utilization in the normal, conscious rhesus monkey.

Methods

Chemicals

2-Deoxy-D-[1-^{14}C] glucose, specific activity 50 to 56 mCi per millimole, was obtained from New England Nuclear Corporation, Boston, MA. Enzymes and reagents used in the assay of blood and plasma glucose concentrations were purchased in kit form from Calbiochem, Richmond, CA. Calibrated [^{14}C] toluene, used for internal standardization of radioactive samples counted in the liquid scintillation counter, was obtained from Packard Instrument Company, Downers Grove, IL, and New England Nuclear Corporation.

Animals

All the studies reported here were carried out on young macaque monkeys weighing between 2.9 and 4.0 kg. They were prepared for the experiments by insertion of polyethylene catheters in a femoral artery and vein under light halothane and nitrous oxide anesthesia. The procedure to determine the lumped constants (described later) required the sampling of representative cerebral venous blood; in these experiments a third polyethylene catheter was introduced into the internal jugular vein and threaded upward to the superior bulb. The animals were placed in a restraining chair and allowed to recover from the anesthesia. The experimental procedures for determining the lumped constant and for measuring local cerebral glucose utilization were carried out two to five hours after recovery from anesthesia with the animals still in the restraining chair.

Miscellaneous Physiological and Biochemical Measurements

Mean arterial blood pressure was measured by means of an air-damped mercury manometer attached to the femoral artery catheter. Hematocrit was determined from blood samples centrifuged in a Beckman Microfuge B (Beckman Instruments, Fullerton, CA), and arterial pH, P_{CO_2}, and P_{O_2} were measured in an ILS Model 213 pH blood gas analyzer (Instrumentation Laboratory, Inc, Lexington, MA).

These physiological variables are not essential to determination of either the lumped constant or local cerebral glucose utilization; they were measured only to establish that the animals were in a normal physiological state. The normal means and standard deviations of these variables established for the conscious monkey in our laboratory are as follows: mean arterial blood pressure, 107 ± 10 torr; hematocrit, 43 ± 3%; arterial pH, 7.50 ± 0.04; arterial P_{CO_2}, 34 ± 4 torr; and arterial P_{O_2}, 83 ± 4 torr. In all the animals used in the present studies, these variables were within 2 standard deviations from the mean.

Blood glucose concentration was measured enzymatically in barium-zinc filtrates [15] by means of reduction of nicotinamide-adenine dinucleotide catalyzed by coupled glucose-dependent hexokinase–glucose-6-phosphate dehydrogenase reactions [21]. Plasma glucose concentration was measured directly in plasma samples either by the same enzymatic procedure or by means of a Beckman glucose analyzer. Plasma and blood [^{14}C]DG concentrations were determined by assay of the ^{14}C content in measured volumes of plasma and barium-zinc filtrates of whole blood, respectively, in a liquid scintillation counter. The phosphor solution consisted of 125 gm of naphthalene, 12 gm of 2,5-diphenyloxazole, and 0.3 gm of *p*-bis-[2-(-5-phenyloxazolyl)]-benzene per liter of *p*-dioxane. The number of millimicrocuries per sample was computed from its counting rate on the basis of internal standardization with a calibrated [^{14}C] toluene standard.

Measurement of Tissue ^{14}C Concentration

The ^{14}C concentrations in localized regions of the brain were measured by a quantitative autoradiographic technique [18] modified to improve resolution and to minimize errors [11, 23]. The densitometric measurements were made with a Photovolt Model 520-A densitometer (Photovolt Corporation, New York, NY) equipped with a 0.2 mm aperture.

The structures selected for determination of rates of glucose consumption were considered to be reasonably representative of all the major subdivisions of gray and white matter of the mature primate brain. In such structures as the thalamus, with its large number of constituent nuclear masses, only a few were selected, and in some cases the choice was influenced mainly by the ability to identify the structure on the basis of its geographic relationship to other, clearly defined regions. The number of measurements of optical density made for any one structure

varied according to its size. In the case of the relatively large caudate nucleus, for example, readings were made through sections of the tail, body, and head. For each structure, readings were made bilaterally and in three serial sections whenever possible. Thus, for large structures, as many as 30 readings were made. For each structure the mean value of all readings was used for calculation of glucose utilization in that structure. For relatively small regions, such as the cochlear nucleus, as few as 6 readings were averaged. In a few regions, such as the auditory or visual cortex, variable optical density was often created either by columns or stripes running perpendicular to the cortical surface or by parallel laminae such as the prominent dark band corresponding to layer IV. Here too, a number of readings were made, even within a small zone, in order to obtain a mean value that could be considered representative of that anatomical region.

Determination of the Lumped Constant

The method for determination of the lumped constant was based upon equations 38 and 40 of our original report of the [^{14}C]DG method [23]. These equations can be rewritten as follows:

$$\left[\frac{C_A^*}{C_A}\right]\left[\frac{(C_A^* - C_V^*)/C_A^*}{(C_A - C_V)/C_A}\right] = \frac{k_1^* C_P^* e^{-(k_2^* + k_3^*)T}}{R_i} + \left[\frac{C_P^*}{C_P}\right]\left[1 - e^{-(k_2^* + k_3^*)T}\right]\left[\frac{\lambda V_m^* K_m}{\Phi V_m K_m^*}\right] \quad (2)$$

where C_A^*, C_V^*, and C_P^* equal the concentrations of [^{14}C]DG in arterial blood, cerebral venous blood, and arterial plasma, respectively; C_A, C_V, and C_P are the concentrations of glucose in arterial blood, cerebral venous blood, and arterial plasma, respectively; C_P^*/C_P remains constant; and the other symbols are the same as those defined for equation 1.

As T approaches infinity, all the terms containing the exponential factor approach zero, and at $T = \infty$ the equation is reduced to:

$$\left[\frac{C_A^*/C_A}{C_P^*/C_P}\right]\left[\frac{(C_A^* - C_V^*)/C_A^*}{(C_A - C_V)/C_A}\right](\infty) = \left[\frac{\lambda V_m^* K_m}{\Phi V_m K_m^*}\right] = \text{lumped constant} \quad (3)$$

These equations state that if [^{14}C]DG is so administered intravenously that the arterial plasma concentration of [^{14}C]DG reaches and maintains a constant level for a long enough period, then the ratio of cerebral extractions of [^{14}C]DG and glucose from the blood multiplied by the ratio of arterial whole blood specific activity (i.e., the ratio of [^{14}C]DG to glucose concentrations) to that of plasma will decline exponentially, with a rate constant equal to $(k_2^* + k_3^*)$, until it reaches an asymptotic level equal to the lumped constant.

The procedures used to apply these equations to determination of the lumped constant in the monkey were similar to those used previously to determine the lumped constant in the rat [23]. A number of monkeys were administered an intravenous pulse of 100 μCi of [^{14}C]DG per kilogram of body weight, and their plasma disappearance curves of [^{14}C]DG were measured, as already described, for 35 to 45 minutes. The plasma disappearance curves were then fitted by a nonlinear, iterative, least-squares routine to a sum of four exponential terms [17]. From a Laplace transformation of this function, an infusion schedule designed to produce a constant arterial plasma concentration was computed [17]. This infusion schedule was used in another group of conscious monkeys that had been prepared for experiments to determine the lumped constant by prior insertion of catheters into a femoral artery and vein and into the superior bulb of the internal jugular vein.

Immediately following an initial intravenous pulse of 10 μCi of [^{14}C]DG, the programmed infusion was instituted by means of a calibrated peristaltic pump equipped with a manual speed control. The speed control was then adjusted by hand every 30 seconds in accordance with the prescribed infusion schedule for a period of 35 to 45 minutes. During the infusion, timed arterial and cerebral venous blood samples of about 100 μl each were drawn from the femoral artery and superior bulb of the internal jugular vein. Part of each blood sample was immediately centrifuged in the Beckman Microfuge to separate the plasma, and the blood and plasma samples were assayed for their [^{14}C]DG and glucose concentrations as described (Fig 1A). From the measured variables in blood and plasma, the lumped constant was calculated by equation 3. The function on the left side of equation 3 was plotted against time, and after it reached a constant or asymptotic level, which was usually from 25 to 30 minutes onward, the values for the function were averaged to obtain the lumped constant (Fig 1B).

In a few animals, satisfactorily constant arterial plasma concentrations of [^{14}C]DG or glucose were not achieved. All experiments in which the arterial plasma concentrations of either compound varied by more than 10% after the first 5 minutes of infusion were rejected from the study.

Determination of Rate Constants

In the original development of the [^{14}C]DG method, which was carried out in rats, the rate constants k_1^*, k_2^*, and k_3^* were determined by a complex procedure that required a large number of matched animals. An equation was derived, equation 11 in our earlier report [23], which defined the tissue concentration of total ^{14}C as a function of time, of the history of the plasma concentration of [^{14}C]DG, and of the three rate constants. A large group of normal male adult rats, matched for age and weight, were administered continuous intravenous infusions of [^{14}C]DG for periods varying from 5 to 40 minutes, during which time samples of arterial blood were taken for determination of the time courses of the arterial plasma [^{14}C]DG concentrations. The animals were killed at the end of the infusion periods, and the ^{14}C concentrations in individual structures of the brain were measured by the quantitative autoradiographic technique. The equation was then fitted to the measured data by a nonlinear, iterative, least-squares routine to obtain the

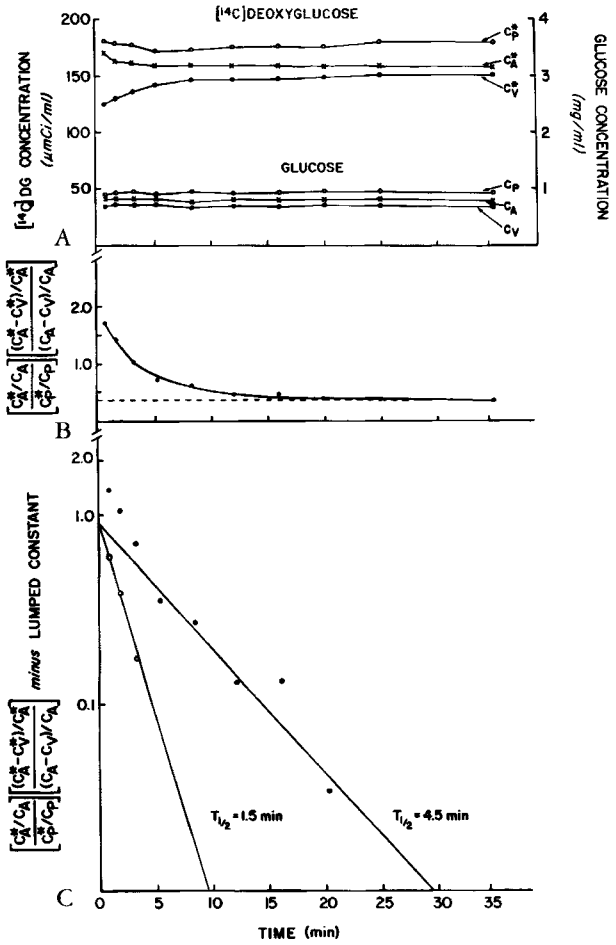


Fig 1. Data obtained and their use in determination of the lumped constant and the combination of rate constants, (k_2^* + k_3^*), in a representative experiment (see Methods). (A) Time courses of arterial blood and plasma concentrations of [^{14}C]*DG and glucose and cerebral venous blood concentrations of* [^{14}C]*DG and glucose during programmed intravenous infusion of* [^{14}C]*DG. (B) Arithmetic plot of the function derived from the variables in A and combined as indicated in the formula on the ordinate against time. This function declines exponentially, with a rate constant equal to (k_2^* + k_3^*), until it reaches an asymptotic value equal to the lumped constant, 0.35, in this experiment* (dashed line).

(C) Semilogarithmic plot of the curve in B less the lumped constant, i.e., its asymptotic value. Solid circles represent actual values. This curve is analyzed into two components by a standard curve-peeling technique to yield the two straight lines representing the separate components. Open circles are points for the fast component, obtained by subtracting the values for the slow component from the solid circles. The rate constants for these two components represent the values of (k_2^* + k_3^*) for two compartments; the fast and slow compartments are assumed to represent gray and white matter, respectively. In this experiment the values for (k_2^* + k_3^*) were found to equal 0.462 (half-time = 1.5 min) and 0.154 (half-time = 4.5 min) in gray and white matter, respectively.

best-fitting values for k_1^*, k_2^*, and k_3^* and their standard errors of estimates for most of the gray and white structures of the rat brain [23]. Because each of the rate constants varied relatively little from one gray or white structure to another, they were averaged for the two types of tissue to obtain a single set of rate constants for each one [23].

The expense and impracticality of obtaining the large numbers of matched animals required by this procedure precluded its use to determine rate constants in the monkey. An alternative method was therefore selected. In equation 1, the operational equation of the [^{14}C]DG method, the rate constants k_2^* and k_3^* always appear as a sum, never separately. Therefore, only their sum is required for solution of this equation, and they do not have to be determined separately. Equations 2 and 3 suggest a means for determination of (k_2^* + k_3^*) in the same experiments carried out to arrive at the lumped constant. These equations state that the quantity

$$\left[\frac{C_A^*/C_A}{C_P^*/C_P}\right]\left[\frac{(C_A^* - C_V^*)/C_A^*}{(C_A - C_V)/C_A}\right]$$

declines monoexponentially, with a rate constant equal to (k_2^* + k_3^*), until it reaches an asymptotic value equal to the lumped constant. Therefore, from data obtained in the experiments to establish the lumped constant, it was possible to determine (k_2^* + k_3^*) from the slope of the semilogarithmic plot of this quantity against time. Inasmuch as cerebral venous blood drains both gray and white matter, which have different values for (k_2^* + k_3^*), the declining curves really represent a sum of two exponential terms with two values for the rate constant (k_2^* + k_3^*), one each for gray matter and white matter. By curve-peeling or computer-fitting techniques, it was possible in some experiments to determine the values for (k_2^* + k_3^*) for gray matter and white matter separately (Fig 1C).

This procedure did not provide the means to determine k_1^* directly. The tissue distribution space for [^{14}C]DG is equal to $k_1^*/(k_2^*$ + $k_3^*)$ [23], and once (k_2^* + k_3^*) is known, k_1^* can be calculated by multiplication of the distribution space by (k_2^* + k_3^*). In order to calculate k_1^*, it had been planned to assume that the distribution space per gram of gray matter or white matter was the same in the monkey as in the rat. For reasons presented in Results, this calculation was unnecessary, and the k_1^* value determined in the rat was used directly in the monkey.

Measurement of Local Cerebral Glucose Utilization

Local cerebral glucose utilization was measured in normal, conscious monkeys sitting in a restraining chair at least two hours after they had recovered from the anesthesia used for insertion of the femoral arterial and venous catheters. Particular attention was given to maintaining relative uniformity of ambient visual and auditory stimulation. The experimental procedure was initiated with an intravenous pulse of 100 μCi of [^{14}C]DG per kilogram of body weight contained in 3 ml of physiological saline solution. Timed arterial samples were then collected directly into heparinized Microfuge tubes at approximately the following times after the onset of the [^{14}C]DG administration: 20

seconds, 40 seconds, and 1, 1.5, 3, 5, 7.5, 15, 25, 35, and 45 minutes. Immediately after the last sample, the animal was killed by rapid intravenous injections of, first, 50 mg of sodium thiopental in 5 ml of saline, and next a saturated potassium chloride solution to stop the heart. The brain was then removed as rapidly as possible and frozen in Freon XII chilled to −50°C with liquid nitrogen. Lower temperatures often resulted in cracking of the brain during freezing. After at least 15 minutes had been allowed for complete freezing, the brain was removed from the cooling bath and stored at −70°C until sectioned for autoradiography. Sectioning, autoradiography, and densitometric analyses of the autoradiographs were carried out as previously described [23]. The rates of local cerebral glucose utilization were calculated according to equation 1 by means of a Hewlett-Packard Model 9830B programmable calculator (Hewlett-Packard Company, Loveland, CO).

Results

Evaluation of Lumped Constant

Experiments to determine the lumped constant were successfully carried out in 7 conscious monkeys. The nature of the data obtained and their use to determine the lumped constant are illustrated by the results of one of these experiments graphically depicted in Figure 1A and B. The mean value of the lumped constant obtained in the 7 experiments was 0.344 (SEM, ± 0.036). This was the value used in determination of local cerebral glucose utilization in the monkey.

Evaluation of Rate Constants

The method for determination of $(k_2^* + k_3^*)$ for gray and white matter from the same data obtained for determination of the lumped constant is illustrated in Figure 1B and C. In 2 of the 7 experiments, the arterial plasma [^{14}C]DG concentration curve was sufficiently constant from zero time onward, and an adequate number of early blood samples could be collected, to permit compartmental analysis of the curve in Figure 1B, the procedure necessary to obtain the values for $(k_2^* + k_3^*)$ for both gray matter and white matter (see Fig 1C). The fast component was assumed to represent gray matter and the slow component, white matter. The means of the values for $(k_2^* + k_3^*)$ for gray matter and white matter obtained in these two experiments were 0.347 and 0.144 per minute, respectively. These values of $(k_2^* + k_3^*)$ are equivalent to half-lives of 2.0 and 4.8 minutes for the free [^{14}C]DG pool in gray matter and white matter, respectively. In an additional experiment it was not possible to separate the fast and slow components from the semilogarithmic plot, but a single component, probably representing a mixture of gray and white matter, with a value for $(k_2^* + k_3^*)$ of 0.217 per minute (half-life = 3.2 min) was found.

These values for $(k_2^* + k_3^*)$ were so similar to those obtained in extensive studies in the rat [23] that it was decided to use those values determined in the rat for measurements of local cerebral glucose utilization in the monkey. Furthermore, because the values for $(k_2^* + k_3^*)$ were so similar in the two species, it seemed reasonable to assume that k_1^* would also be similar, and the values for k_1^* measured in the rat were used in the studies in the monkey. The values of k_1^*, k_2^*, and k_3^* used in the present studies were therefore 0.189, 0.245, and 0.052 in gray matter and 0.079, 0.133, and 0.020 in white matter, respectively. It should be noted that even if the values for rate constants in the rat do not apply exactly to the monkey, the error to be expected in the final values for local glucose utilization will be relatively small. The use of a pulse of [^{14}C]DG followed by an interval of 45 minutes before killing, the procedure used in the [^{14}C]DG method, serves to minimize the significance of the terms in equation 1 that contain the rate constants, and therefore also the errors that might result from inaccuracies in their values [23].

Local Cerebral Glucose Utilization

The values for the rates of glucose utilization in 31 gray and 8 white structures in the brain of the conscious monkey are listed in the Table. The rates in white matter were very low and were relatively uniform from one structure to another. The rates in gray matter were quite heterogeneous but almost uniformly much higher than those in white matter. The highest value was in the inferior colliculus, as has also been observed in the rat [23]. Not only was there considerable variability among different gray structures, but in many cases there was heterogeneity within defined anatomical structures (Figs 2, 3). This heterogeneity did not always correspond to cytoarchitectural features revealed by traditional histological techniques. For example, columnar distribution of alternating higher and lower rates of glucose utilization were seen to some degree in most regions of the cerebral cortex (Figs 2A,C,D). A thin dorsal layer of superior colliculus exhibited a rate of glucose utilization approximately 50% greater than the mean of the entire structure (Fig 3B). In the cerebellar vermis, linear bands perpendicular to the surface of the folia marked discrete regions of higher rates of glucose utilization (Fig 3C). The values presented in the Table represent the average values in the structures listed although it should be recognized that there are differences even within these structures.

Discussion

To our knowledge, the present studies represent the first published report of quantitative determination of rates of cerebral energy metabolism in the normal, conscious monkey, not only at the local level, but

Values for Local Cerebral Glucose Utilization in Normal, Conscious Monkeys

Structure	Value (μmoles/ 100 gm/min)
Gray matter	
Visual cortex	59 ± 2
Auditory cortex	79 ± 4
Parietal cortex	47 ± 4
Motor cortex	44 ± 3
Frontal cortex	50 ± 2
Thalamus: dorsomedial nucleus	54 ± 2
Thalamus: ventral posterolateral nucleus	43 ± 2
Medial geniculate body	65 ± 3
Lateral geniculate body	39 ± 1
Hypothalamus: ventromedial nucleus	25 ± 1
Mamillary body	57 ± 3
Hippocampal gyrus	39 ± 2
Amygdala: anterior nucleus	25 ± 2
Septal nucleus: medial	26 ± 2
Caudate nucleus	52 ± 3
Nucleus accumbens	36 ± 2
Putamen	53 ± 3
Globus pallidus	26 ± 2
Subthalamic nucleus	51 ± 2
Substantia nigra: zona diffusa	29 ± 2
Vestibular nucleus	66 ± 3
Cochlear nucleus	51 ± 3
Superior olivary nucleus	63 ± 4
Inferior colliculus	103 ± 6
Superior colliculus	55 ± 4
Pontine reticular nucleus	28 ± 1
Vermis	32 ± 2
Flocculus	50 ± 3
Cerebellar cortex	31 ± 2
Cerebellar nuclei	45 ± 2
Spinal gray matter: cervical[a]	21 ± 2
White matter	
Optic chiasm[a]	23 ± 2
Optic tract	18 ± 1
Corona radiata	11 ± 1
Corpus callosum	11 ± 0
Internal capsule	13 ± 1
Pyramids	13 ± 1
Cerebellar white matter	12 ± 1
Spinal white matter[a]	7 ± 1

The values are means ± standard errors from measurements in 7 animals except those marked with [a], which represent values in 6 animals.

even in the brain as a whole. There have, however, been measurements of cerebral energy metabolism in the anesthetized monkey. Schmidt, Kety, and Pennes [20] reported a cerebral oxygen consumption of 3.7 cc per 100 gm of tissue per minute in the monkey under barbiturate anesthesia, but the level of anesthesia was kept very low to approach and approximate the conscious state. If one assumes the usual stoichiometry between the rates of cerebral oxygen and glucose utilization, then this rate is equivalent to approximately 28 μmoles of glucose per 100 gm of tissue per minute, a rate very similar to that of 31 μmoles per 100 gm per minute observed in normal conscious young men [22]. The rates of glucose utilization obtained in the present studies are for individual components of the brain. To derive the average rate in the brain as a whole from these data, it would be necessary to know the appropriate mass-weighting factors for the various structures, and these factors are not available for the monkey. It is reassuring, however, that the rates for the individual tissues bracket the previously observed average value for the brain as a whole. Indeed, if the monkey brain is estimated to consist of equal fractions of gray and white matter, and the arithmetic means of the rates in the gray structures and in the white structures are averaged on that basis, then that average is found to be 29 μmoles per 100 gm of tissue per minute, very close to the values of 28 and 31 μmoles per 100 gm per minute previously observed in the lightly anesthetized monkey [20] and in conscious man [22].

As in the rat [23], the rates of local cerebral glucose utilization in the monkey vary over a wide range and fall into two distinct distributions, a relatively broad one for gray matter and a fairly narrow one for white matter. The values in gray matter are generally three to five times those in white matter. Also, as in the rat, the components of the auditory system are metabolically the most active structures of the brain with the highest rate of glucose utilization in the inferior colliculus.

These values for local cerebral glucose utilization in the conscious monkey are considerably below those observed in the conscious rat [23]. The high rate of cerebral energy metabolism in the rat has previously been shown in measurements in the brain as a whole [4, 5, 16]. The present results demonstrate that the high average rate in rat brain is not merely because of the lesser content of white matter; even equivalent gray structures have higher rates in the rat than in the monkey. A likely explanation is the higher cellular packing density in rat brain [16].

The values for rate of glucose utilization in the individual structures listed in the Table are average values for those structures. Not unexpectedly, however, there was considerable heterogeneity in many of the structures, which in some cases corresponded to their known histological cytoarchitecture. For example, a dark lamina was clearly discernible in the cortical mantle of area 17 (see Fig 2A), which has been shown to correspond to layer IV [11]. It is the metabolism of this layer that is most prominently affected by alterations in visual input [11]. Heterogeneity was also observed in other cortical regions, usually in the form of columns of variable width running perpendicular to the surface. In area 18, these col-

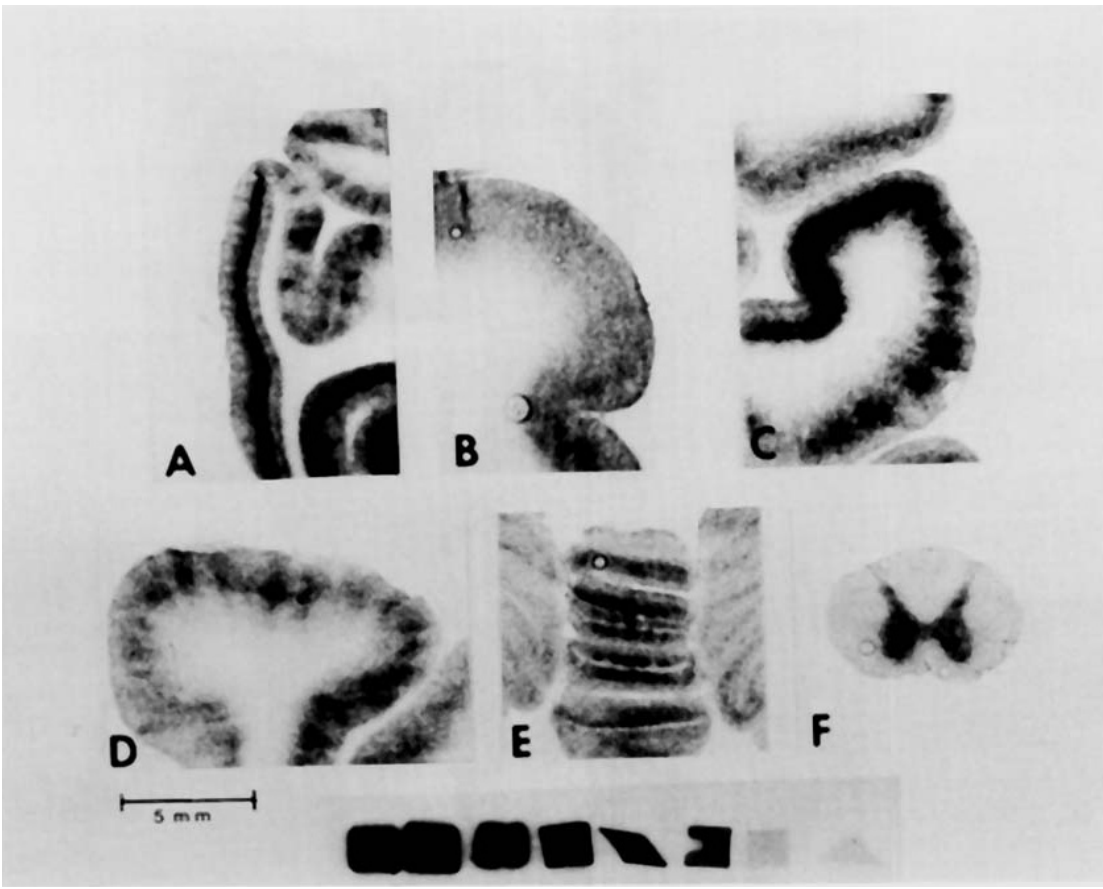


Fig 2. Autoradiographs of selected regions of the cerebral cortex, the uvula of the cerebellum, and the spinal cord. The autoradiographs of the plastic ^{14}C standards are also visualized.

(A) Visual cortex—parasagittal section at the junction of areas 17 and 18. The dark, centrally located band running parallel to the surface (left) marks layer IV in area 17. At its site of abrupt termination there is a corresponding change in cytoarchitecture to indicate the transition to area 18 [11]. Note the irregularly spaced markings, which are perpendicular to the surface and which are uniformly narrow in area 17 but are more variable in width in area 18.

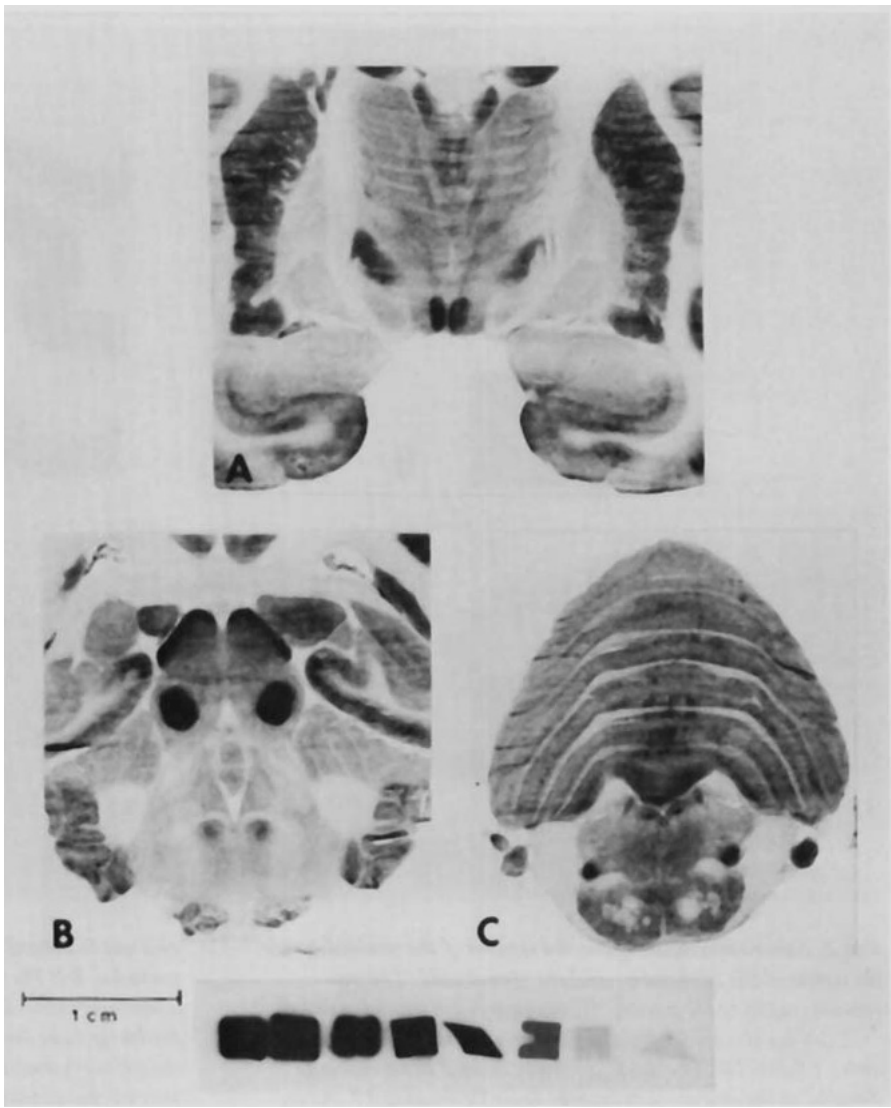
(B) Motor cortex. In most of this cortex, radial streaks or columns are not seen. This may be indicative of the animals' relative inactivity during the study.

(C) Auditory cortex—coronal section through the superior temporal gyrus. No matching histological examination was carried out to identify by cytoarchitecture which subdivisions are included, but the section shows both laminar (dorsomedial) and columnar (ventrolateral) patterns. Such patterns of neuronal discharge have been observed in response to pure tone stimuli in the primary auditory cortex [1, 13]. The rate of glucose utilization in the centrally placed dark lamina is 121 μmoles per 100 gm of tissue per minute. In the streaks on the lateral surface it varies from 47 to 67 μmoles per 100 gm per minute.

(D) Frontal cortex. Sporadically located radial streaks mark the entire frontal cortex.

(E) Uvula of the cerebellum. Bands traversing the folia are regularly spaced.

(F) Spinal cord at the level of the first cervical vertebra. The difference in metabolic rates clearly delineates gray and white matter.



umns measure only about 100 to 200 μm (see Fig 2A). In all cortical regions they traverse the entire cortical thickness although in some zones they appear as spots because of the plane of sectioning. These columns are distinctly different from the ocular dominance and orientation columns, which are different in nature and appearance, confined to the striate cortex, and elicited only by specific experimental conditions [7–9, 11]. Whether the columns seen in these studies reflect some undefined and unrecognized experimental conditions associated with the present studies is uncertain, but the columnar pattern may represent the metabolic counterpart of the general organization of the cortex revealed by electrophysiological and anatomical techniques [1, 2, 7–10, 14]. It may also be related to the columnar distribution of adenosine triphosphate that has recently been demonstrated in the cerebral cortex by histochemical techniques [12].

Striking heterogeneity was also seen in structures

Fig 3. Autoradiographs of miscellaneous coronal sections of monkey brain and calibrated plastic ^{14}C standards used for quantification of autoradiographs. (A) Diencephalon. Relatively high metabolic rates are marked by dark regions in the mamillary nuclei, the subthalamus, and the putamen. (B) Midbrain. The inferior colliculi have the highest metabolic rate of all structures in the brain, and this rate is relatively uniform throughout the structure. This is in contrast to the superior colliculus, where only the most superficial portion has a high rate. In this animal the glucose utilization rate in the superficial layer was 107 μmoles per 100 gm of tissue per minute, 67% above that of the deep portion of the structure. (C) The anterior cerebellum and pons at the level of the superior olives. Note the dark streaks running perpendicular to the folia in the vermis and the very dark lingula.

other than the cerebral cortex. The extreme superficial layer of the superior colliculus was found to have a metabolic rate usually 20 to 30% above that of the deeper portion (see Fig 3B), a finding consistent with electrophysiological observations [6, 19]. The scattered linear streaks seen in the vermis of the cerebellum (see Fig 3C) have, to our knowledge, no counterpart in electrophysiological and anatomical studies. They may represent selected loci of activity driven by afferent stimuli from the spino-olivocerebellar pathway [3]. From observations of the internal distributions of metabolic activity within the structures, it appears that the deoxyglucose method may offer, with or without quantification, a powerful new tool with which to study the functional organization of the central nervous system.

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